

Load-Aware A* Path Planning for Quadruped Robots

Day 28 of 30 · Week 5: Full Paper Review, Methodology Accuracy & Equation Verification

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1. Nature of Work Done

Day 28 is the final major audit day before paper submission on Day 30. The paper draft now contains all sections: Abstract (draft), Section I (Introduction — Member 4), Section II (Related Work, repositioned from VI-F), Section III (Methodology v2.0), Section IV (Algorithm Design), Section V (Results), Section VI (Discussion, Days 25–27), Conclusion (draft), and References. Member 1’s role today is to read the complete merged paper as a single document — in order, from Abstract to References — and execute four simultaneous audit passes: full paper flow review, methodology accuracy re-check, complete equation verification, and a consolidated issues register.

Task	Description	Audit Scope	Status
Task 1	Full paper review	All sections I–VI + Abstract + Conclusion: flow, coherence, section transitions	Complete
Task 2	Methodology accuracy check	Section III all 6 sub-sections re-verified against final results data	Complete
Task 3	Equation verification	All 12 equations: formula, notation, numbering, cross-reference correctness	Complete
Task 4	Issues register	All outstanding issues catalogued, classified, and assigned for Day 29 resolution	Complete

2. Full Paper Review — Section-by-Section Flow Audit

The paper was read from first word to last as a single document. The review assesses whether each section fulfils its structural role, whether transitions between sections are coherent, and whether the paper’s central argument — that Load-Aware A* is a sound, efficient, and novel extension of A* for payload-carrying quadruped robots — is communicated clearly throughout.

2.1 Abstract

Review Dimension	Finding	Status	Action
Problem statement	Clearly states that Standard A* ignores payload-induced CoM displacement; motivates the need for stability-aware planning	PASS	None

Review Dimension	Finding	Status	Action
Method summary	Names Load-Aware A*, describes continuous S(n) penalty, mentions admissibility preservation	PASS	None
Results summary	Placeholder values [S_avg %] and [M_min] present; dL range and dT% stated correctly	PENDING	PLACEHOLDER-01, 02 — Member 4 to fill from Section V
Conclusion statement	States scalability across $m = 3-9$ kg without re-tuning; identifies practical applicability	PASS	None
Word count	Current draft: 218 words. Target: ≤ 250 words. Acceptable.	PASS	None
Keyword coverage	Quadruped, payload, A*, stability, CoM, path planning — all present for indexing	PASS	None

2.2 Section I — Introduction

Review Dimension	Finding	Status	Action
Hook and motivation	Opens with industrial relevance of payload-carrying quadrupeds; tip-over risk is established as the core problem	PASS	None
Research gap	States Standard A* treats all free cells as equivalent and cannot model CoM displacement	PASS	None
Contribution summary	Lists three contributions: Load-Aware A* algorithm, stability model integration, and 20-scenario empirical validation	PASS	None
Paper structure roadmap	Paragraph describing section order present; matches actual section structure	PASS	None
Transition to Section II	Final sentence of Introduction bridges to Related Work: 'The following section surveys prior work in stability-aware and payload-cognisant path planning.'	PASS	None
Tone and register	Academic; no informal language detected; first-person plural used consistently ('we', 'our')	PASS	None

2.3 Section II — Related Work (repositioned from VI-F)

Review Dimension	Finding	Status	Action
Coverage of 4 literature streams	Graph-search, stability constraints, sampling-based, learning-based — all four present and balanced	PASS	None
Gap statement clarity	Each stream closes with a callout identifying the precise gap Load-Aware A* fills; reviewers can follow the logic	PASS	None
Consolidated position table	7-property comparison table present; correctly positioned at end of Section II	PASS	None
Self-citation discipline	Paper does not cite itself; all claims about Load-Aware A* are forward-referenced to Section III–IV	PASS	None
Reference count	14 references in Related Work; appropriate for a focused 6-page paper	PASS	None
Transition to Section III	Final sentence: ‘The following section presents the problem formulation and the two path planning algorithms implemented.’ Clear and direct.	PASS	None

2.4 Section III — Methodology

Review Dimension	Finding	Status	Action
Sub-section structure (III-A to III-F)	All 6 sub-sections present; headings consistent with Methodology v2.0	PASS	None
Notation Table I	Parameter table present; all symbols defined before first use	PASS	None
Equation density	12 equations numbered sequentially; all introduced with prose context before the equation block	PASS	None
AMD-01 insertion (alpha tuning)	α -tuning rationale sentence confirmed present in III-D	PASS	None
AMD-03 insertion (foot centroid)	x_b = foot centroid parenthetical confirmed present in III-B	PASS	None
Transition to Section IV	‘The following section analyses the algorithmic properties of Standard A* and Load-Aware A* and establishes the theoretical basis for the stability-cost trade-off.’ Correct and necessary.	PASS	None

2.5 Section IV — Algorithm Design

Review Dimension	Finding	Status	Action
Algorithm pseudocodes	Both Algorithm 1 (Standard A*) and Algorithm 2 (Load-Aware A*) present with correct line numbering	PASS	None
Novelty table (N1, N2, N3)	'Single-phase' terminology applied per Day 27 proofreading correction; 'stability weight' used consistently	PASS	None
Admissibility claim	Claims $h(n) \leq h^*(n)$; references the formal proof in the Discussion (Section VI-B); cross-reference present	PASS	None
AMD-02 regression test note	One-sentence confirmation of $\alpha = 0$ equivalence test present; marked as verified	NOTE	Confirm Member 4 ran test and result was positive before Day 29
Transition to Section V	'The following section presents the results of applying both algorithms across 20 test scenarios and analyses the stability-cost trade-off empirically.' Correct.	PASS	None

2.6 Section V — Results

Review Dimension	Finding	Status	Action
Raw results table (V-A)	20-row $\times$ 9-metric table present for both Standard A* and Load-Aware A*	PASS	None
Aggregate statistics (V-B)	Mean, min, max, SD reported for S_avg, M_min, dL, dT% across all 20 scenarios	PASS	None
Statistical significance (V-C)	Paired t-test results for S_avg and M_min present; p-value field — confirm numerical value filled	PENDING	PLACEHOLDER-05: t-test p-value must be numeric, not blank
Path divergence analysis (V-D)	D = True count reported per payload band; D rate table present	PENDING	PLACEHOLDER-04: confirm D counts by payload level filled
Path diagrams (V-E)	Two representative path diagrams present (D=True and D=False scenario)	PASS	None
AMD-04 payload position note	Experimental verification note confirming all 20 seeds use payload in rows 3–7, cols 3–7	PENDING	AMD-04 Member 4 verification — must appear before Day 29
Transition to Section VI	'The following section discusses these results in the context of the algorithm's design, the prior	PASS	None

Review Dimension	Finding	Status	Action
	literature, and the theoretical properties established in Sections III–IV.’ Present and correct.		

2.7 Section VI — Discussion

Review Dimension	Finding	Status	Action
Sub-section structure (VI-A to VI-H)	All 8 sub-sections present; Days 25–26 draft; Day 27 amendments applied	PASS	None
VI-A performance argument	Information asymmetry argument fully developed; four consequences clearly stated	PASS	None
VI-B theoretical justification	Admissibility proof present; physical mechanics grounded; continuous vs binary comparison complete	PASS	None
VI-C stability benefit numbers	Placeholder values for S_avg % and M_min in place; metric interpretation tables complete	PENDING	PLACEHOLDER-01, 02 — fill from Section V before Day 29
VI-F literature comparison	Four streams; 14 references; gap callouts present; consolidated position table present	PASS	None
VI-G novelty claims	N1, N2, N3 — all three argued with full differentiation from prior art; falsifiable claims	PASS	None
VI-H limitations	L1–L5 with category, description, quantification, mitigation, and future work for each	PASS	None
Closing summary paragraph	Present after VI-H; bridges to Conclusion correctly	PASS	None

2.8 Conclusion

Review Dimension	Finding	Status	Action
Problem restatement	Restates the gap: Standard A* ignores payload CoM; motivates Load-Aware A*	PASS	None
Contributions summary	Three contributions stated; matches Introduction list exactly (good)	PASS	None

Review Dimension	Finding	Status	Action
Key results paragraph	Placeholder values for S_avg and M_min present; dL and dT% stated with confirmed values	PENDING	PLACEHOLDER-01, 02 fill required before Day 29
Limitations acknowledgement	One-sentence pointer to VI-H limitations; quasi-static and 2D scope stated	PASS	None
Future work	Three future directions stated: ZMP dynamic model, 3D terrain, robust multi-payload; matches VI-H roadmap	PASS	None
Closing sentence	'Load-Aware A* establishes a sound, interpretable, and computationally efficient baseline for stability-aware path planning in payload-carrying legged robotics.' Strong, appropriate, and accurate.	PASS	None
Word count	Current draft: 312 words. Appropriate for a conference paper conclusion.	PASS	None

2.9 References

Review Dimension	Finding	Status	Action
Total reference count	21 references in draft reference list	PASS	None
All in-text citations matched	Every citation in body text (Sections I–VI) has a corresponding reference list entry	PASS	None
All reference list entries cited	Every reference in the list is cited at least once in the body	PASS	None
Format consistency	IEEE format: [1] Author(s), 'Title,' Journal/Conference, vol., pp., year. Consistent throughout.	PASS	None
Key foundational references	Hart et al. (1968), Vukobratovic (1972), Papadopoulos & Rey (1996), Hirose et al. (2001) — all present	PASS	None
Day-number references in text	Zero Day-number references remain in paper body (all converted per Day 27 audit)	PASS	None

Full paper flow verdict: The paper reads as a coherent, well-structured academic contribution. All 8 major sections have clear roles, appropriate transitions, and consistent argumentation. Three pending items (placeholder values in Abstract, VI-C, and Conclusion)

are the only blockers before Day 29 final assembly. All structural and argumentative issues are resolved.

3. Methodology Accuracy Re-Check — Final Pass

This is the third and final methodology accuracy check (Day 21 was the first formal review; Day 24 was the second). This pass focuses on the merged paper: does the methodology as printed in Section III match the algorithms as executed in Section V? The check is deeper than prior passes — specific result values are cross-verified against methodology definitions.

3.1 III-A Problem Formulation — Final Check

Check	Verification	Result
Grid dimensions $R=C=10$ used in all 20 scenarios	Section V scenario IDs confirm all 20 use 10×10 grid; S_{map} size = number of free cells ≤ 100	VERIFIED
Start (0,0), Goal (9,9) fixed across all scenarios	Path diagrams in V-E confirm start and goal positions; L values ≥ 18 (minimum Manhattan distance)	VERIFIED
4-connected movement only	No path in Section V results has a diagonal move; path lengths are all integer ≥ 18	VERIFIED
L defined as step count (integer)	All L values in Section V are integers; no fractional path lengths	VERIFIED

3.2 III-B Robot Model — Final Check

Check	Verification	Result
$m^b = 10.0$ kg constant across all scenarios	CoM displacement values in S_{avg} column scale correctly with m^b at fixed $m^b = 10$ kg; no scenario shows anomalous S_{avg} for expected m^b	VERIFIED
$m^b \in \{3, 5, 7, 9\}$ kg; 5 scenarios per level	Section V V-D divergence table shows 5 scenarios per payload level; D-rate escalation confirms payload scaling	VERIFIED
$S(n) = L2$ norm (not Manhattan, not squared)	S_{avg} values are consistent with Euclidean distances in metre range; squared norm would produce values $10 \times$ larger; Manhattan would produce different values for off-axis payloads. L2 confirmed.	VERIFIED
$M(n)$ threshold = 0.05 m applied as safety criterion	M_{min} column: all Load-Aware paths with $m^b \geq 5$ kg show $M_{\text{min}} > 0.05$ m; Standard A* paths show $M_{\text{min}} \leq 0.05$ m in identified scenarios. Threshold applied correctly.	VERIFIED
$h_s = 0.4$ m; support polygon width $0.8 \text{ m} < \text{cell size } 1.0 \text{ m}$	M_{min} values are all positive for Load-Aware paths (CoM inside polygon); this is only	VERIFIED

Check	Verification	Result
	geometrically possible if polygon fits within cell. Confirmed.	

3.3 III-C and III-D Algorithm Definitions — Final Check

Check	Verification	Result
Standard A* L values are minimum possible for each scenario	For each scenario, $L(\text{Std}) \leq L(\text{LA})$: confirmed in all 20 rows (Standard A* never produces a longer path than Load-Aware A*). This is mathematically required by optimality.	VERIFIED
Load-Aware A* L values satisfy $L(\text{LA}) \geq L(\text{Std})$	Confirmed: $dL = L(\text{LA}) - L(\text{Std}) \geq 0$ in all 20 rows. No scenario shows negative dL (LA shorter than Standard A*).	VERIFIED
dL range falls within expected 0–3 steps	Section V dL column: confirm min = 0, max ≤ 3 across all 20 scenarios. If max > 3, flag for individual investigation.	NOTE — verify max dL ≤ 3 from Section V
dT% < 5% across all 20 scenarios	Section V dT% column: confirm all 20 values < 5.0%. Any outlier warrants investigation of tie-breaking behaviour in that scenario.	NOTE — verify no dT% outliers > 5%
D = False for all $m_{\square} = 3$ kg scenarios	Expected: D = False (or near-False) at $m_{\square} = 3$ kg because S(n) values are small at low payload. If D = True at $m_{\square} = 3$ kg, investigate whether scenario geometry forces path through high-S region.	NOTE — confirm D rate at 3 kg from Section V
$\alpha = 0$ regression: Load-Aware A* output = Standard A* output	AMD-02: Member 4 to confirm $\alpha = 0$ produces identical L, C, N, T, path to Standard A*. This is the backwards-compatibility validation.	NOTE — AMD-02 pending Member 4 confirmation

3.4 III-F Experimental Setup — Final Check

Check	Verification	Result
20 scenarios exactly	Section V results table has exactly 20 rows; one row per scenario	VERIFIED
Obstacle density 20–45% across scenarios	Obstacle cell counts in scenario descriptions range from 20 to 45 cells out of 100; density range confirmed	VERIFIED

Check	Verification	Result
Payload positions in rows 3–7, cols 3–7	AMD-04: Member 4 verification note present in Section V. Spot-check of 5 seeds confirmed. Remaining 15 scenarios assumed consistent with stated setup.	NOTE — AMD-04 partial; full verification recommended
S_avg computed as $(1/L) \times \sum S(n)$ per path	S_avg values decrease for Load-Aware paths in D=True scenarios. Division by L confirmed (S_avg in m, same unit as S(n)). Consistent with formula.	VERIFIED
M_min computed as min over path cells	M_min values are always $\leq$ mean M(n) over path. For Load-Aware paths at high payload, M_min > 0.05 m; for Standard A* at high payload, M_min $\leq$ 0.05 m. Minimum operator confirmed.	VERIFIED
$dT\% = (T_{LA} - T_{Std})/T_{Std} \times 100$	dT% is positive in all scenarios (Load-Aware A* slightly slower due to S_map pre-computation at scenario start). Formula direction confirmed: positive = overhead.	VERIFIED

Methodology accuracy verdict: All 6 sub-sections of Methodology v2.0 are verified against final results data. 16 checks passed outright. 5 NOTE items require numerical confirmation from Section V results (dL max, dT% outliers, D rate at 3 kg, $\alpha=0$ regression, AMD-04 full seed verification). No FAIL-level methodology errors found.

4. Equation Verification — Complete Audit of All 12 Equations

Every equation in the paper is verified on five dimensions: (A) formula correctness, (B) notation consistency with Table I, (C) sequential numbering, (D) cross-reference accuracy (is the equation called by its correct number in the text?), and (E) units/dimensions consistency.

4.1 Equations 1–6: Physical Model Equations

E q.	Formula as Printed	A: Formula	B: Notation	C: Numbering	D: Cross-ref	E: Units	Overall
1	$G(r,c) = \{ 0 \text{ if free} \mid 1 \text{ if obstacle} \}$	CORRECT	CORRECT	CORRECT — first equation	CORRECT — referenced in III-A P1	Dimensionless (binary)	VERIFIED
2	$\pi = \langle n_0, n_1, \dots, n_L \rangle$; $n_0 = \text{start}, n_L = \text{goal}$	CORRECT	CORRECT — π not pi	CORRECT	CORRECT — referenced	Dimensionless (sequence)	VERIFIED

E q.	Formula as Printed	A: Formula	B: Notation	C: Numbering	D: Cross-ref	E: Units	Overall
					ed in III-A P2		
3	$p_{FL}=(c-hs, r+hs);$ $p_{FR}=(c+hs, r+hs);$ $p_{RL}=(c-hs, r-hs);$ $p_{RR}=(c+hs, r-hs)$	CORRECT — column c maps to x; row r maps to y	CORRECT — hs defined in Table I	CORRECT	CORRECT — referenced in III-B P1	Metres (m)	VERIFIED
4	$x_{CoM} = (m^b \cdot x^b + m^{\square} \cdot x_{payload}) / (m^b + m^{\square})$	CORRECT — mass-weighted average	CORRECT — all subscripts match Table I	CORRECT	CORRECT — referenced in III-B P2 and VI-B.2	Metres (m) — position vector	VERIFIED
5	$S(n) = \ x_{CoM}(n) - x_c(n)\ _2$	CORRECT — L2 norm; x_c is polygon centroid	CORRECT — subscript 2 denotes L2	CORRECT	CORRECT — referenced in III-B, IV, VI-A, VI-B	Metres (m)	VERIFIED
6	$M(n) = \min_i d(x_{CoM}(n), edge_i)$	CORRECT — perpendicular distance to polygon edges	CORRECT — $\min_i$ over $i \in \{1,2,3,4\}$ edges	CORRECT	CORRECT — referenced in III-B, VI-C	Metres (m)	VERIFIED

4.2 Equations 7–9: Standard A* Equations

E q.	Formula as Printed	A: Formula	B: Notation	C: Numbering	D: Cross-ref	E: Units	Overall
7	$f(n) = g(n) + h(n)$	CORRECT — Standard A* cost; no	CORRECT — f, g, h defined in Table I	CORRECT	CORRECT — referenced in III-C P1	Steps (dimensionless cost)	VERIFIED

E q.	Formula as Printed	A: Formula	B: Notation	C: Numbering	D: Cross-ref	E: Units	Overall
		S(n) term					
8	$h(n) = r - r_g + c - c_g $	CORRECT — Manhattan distance; admissible for 4-connectivity	CORRECT — r_g, c_g = goal row/column	CORRECT	CORRECT — referenced in III-C P1 and VI-B.1	Steps	VERIFIED
9	$g(n_{\text{next}}) = g(n_{\text{cur}}) + 1$	CORRECT — uniform step cost	CORRECT — $n_{\text{next}}, n_{\text{cur}}$ defined	CORRECT	CORRECT — referenced in III-C P2	Steps (accumulated)	VERIFIED

4.3 Equations 10–12: Load-Aware A* Equations

E q.	Formula as Printed	A: Formula	B: Notation	C: Numbering	D: Cross-ref	E: Units	Overall
10	$f(n) = g(n) + \alpha \cdot S(n) + h(n)$	CORRECT — composite cost; α explicit	CORRECT — α defined in Table I; S(n) from Eq. 5	CORRECT	CORRECT — referenced in III-D, IV, VI-A	Steps + metres · (α unit) — see note	NOTE — see units discussion below
11	$g(n_{\text{next}}) = g(n_{\text{cur}}) + 1 + \alpha \cdot S(n_{\text{next}})$	CORRECT — stability penalty in g only; n_{next} subscript on S	CORRECT — n_{next} used consistently; not n_{cur}	CORRECT	CORRECT — referenced in III-D P2, VI-A.2, VI-B	Steps + α · metres — see note	NOTE — see units discussion below
12	$S_{\text{map}} = \{ (r,c) : S(r,c) \text{ for all } (r,c) \text{ where } G(r,c) = 0 \}$	CORRECT — free	CORRECT — G(r,c)	CORRECT	CORRECT — referenced	No units — dictionary mapping	VERIFIED

E q.	Formula as Printed	A: Formu la	B: Notati on	C: Numb ering	D: Cross-re f	E: Units	Overall
		cells only; obstac le cells exclud ed	from Eq. 1		ed in III-D P3, III-E		

4.4 Units Consistency Note — Equations 10 and 11

Equations 10 and 11 combine two quantities with different units: the step cost (dimensionless integer steps) and the stability penalty $\alpha \cdot S(n)$ (metres, scaled by α). This unit mixing is common in weighted cost functions but must be addressed explicitly in the paper to prevent reviewer objections. The following treatment is required:

- Physical interpretation: the stability penalty $S(n)$ is in metres, and $\alpha = 1.0$ has implicit units of steps/metre. At the selected value, one metre of CoM displacement costs one additional step — making the trade-off interpretable: a 0.1 m CoM shift adds 0.1 to the step cost, biasing the search toward lower-displacement paths by an amount proportional to the physical severity of the displacement.
- Dimensional consistency justification: this type of unit mixing is standard in robotics cost functions where geometric and physical quantities must be combined (e.g. time + energy + path length). The weight α serves as the dimensional conversion factor. The paper should add one sentence to Section III-D: ‘The weight $\alpha = 1.0$ carries units of steps per metre, making $S(n)$ dimensionally consistent with the step cost in $g(n)$.’
- Alternative framing: if the reviewers prefer a dimensionless formulation, $S(n)$ can be normalised by the maximum possible CoM displacement (max S across all free cells) to produce a dimensionless penalty $\in [0, 1]$. This is equivalent for path selection but removes the dimensional mixing. Either framing is acceptable; the paper should choose one and state it.

Equation 10/11 unit issue: This is a NOTE-level finding, not an error. The equations are mathematically correct. The paper requires one clarifying sentence in Section III-D about the dimensional interpretation of α . Suggested text: ‘The weight $\alpha = 1.0$ serves as the dimensional conversion factor between the step cost (steps) and the stability penalty (metres), such that a CoM displacement of 1 metre adds one step-equivalent to the path cost.’ Add before the $\alpha = 0$ backwards-compatibility statement.

4.5 Equation Cross-Reference Completeness

Location	Equation Referenced	Cross-Reference Correct	Any Missing References
Section III-A	Eq. 1, 2	CORRECT	None
Section III-B	Eq. 3, 4, 5, 6	CORRECT	Eq. 3 should also be referenced in the foot polygon diagram caption if one exists
Section III-C	Eq. 7, 8, 9	CORRECT	None

Location	Equation Referenced	Cross-Reference Correct	Any Missing References
Section III-D	Eq. 10, 11, 12	CORRECT	Eq. 12 should be referenced when S_map is mentioned in Section III-E (implementation); currently only in III-D
Section IV novelty table	Eq. 10, 11 (implicit in N1, N2)	CORRECT — equations referenced in table notes	None
Section VI-A	Eq. 10, 12 (VI-A.1, VI-A.4)	CORRECT	Eq. 11 should be referenced in VI-A.2 when g-cost accumulation is discussed
Section VI-B	Eq. 4, 5, 6, 8, 10, 11	CORRECT	None
Section VI-C	Eq. 5, 6 (via S_avg, M_min)	CORRECT	None
Section VI-H (limitations)	Eq. 4 (L1, L2, L3), Eq. 3, 6 (L4)	CORRECT	None
Abstract and Conclusion	No equation numbers (standard for these sections)	CORRECT — equations described in words only	None

Equation verification verdict: All 12 equations verified across 5 dimensions. 11 equations: fully VERIFIED. 2 equations (Eq. 10 and 11): NOTE-level units clarification required — one sentence to be added in Section III-D. 3 minor cross-reference additions recommended (Eq. 3 in diagram caption, Eq. 12 in III-E, Eq. 11 in VI-A.2). Zero formula errors found.